

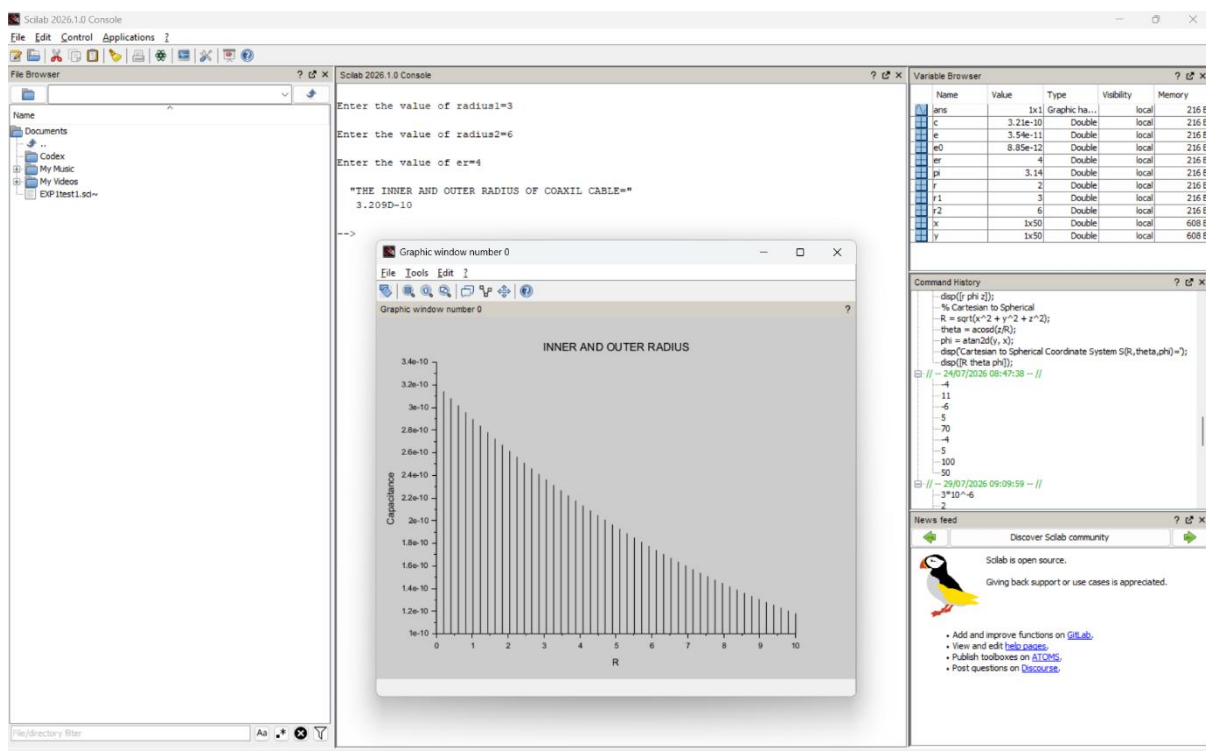
EXP 8: Determination and plotting of Capacitance of Co-axial cable per unit length with different (i) inner radius, (ii) outer radius and (iii) relative permittivity of the dielectric medium.

```

1 clear;
2 clear;
3 r1=input("Enter the value of radius1=");
4 r2=input("Enter the value of radius2=");
5 er=input("Enter the value of er=");
6 r=r2/r1;
7 e0=8.854*10^-12;
8 pi=3.14;
9 e=e0*er;
10 c=(2*pi*e)/(log(r));
11 disp('THE INNER AND OUTER RADIUS OF COAXIAL CABLE=',c);
12 x=linspace(0,10,50);
13 y=c*exp(-0.1*x);
14 figure();
15 subplot(1,1,1);
16 plot(x,y);
17 xlabel('R');
18 ylabel('Capacitance');
19 title('INNER AND OUTER RADIUS');

```

Line 19, Column 32.



exp 8:-

$$r_1 = 3 \text{ mm}$$

$$r_2 = 6 \text{ mm}$$

$$\epsilon_r = 4$$

Capacitance of a co-axial cable

Formula:-

$$C = \frac{2\pi\epsilon}{\ln\left(\frac{r_2}{r_1}\right)}$$

$$C = \frac{2 \times 3.14 \times 35.416 \times 10^{-12}}{0.6931}$$

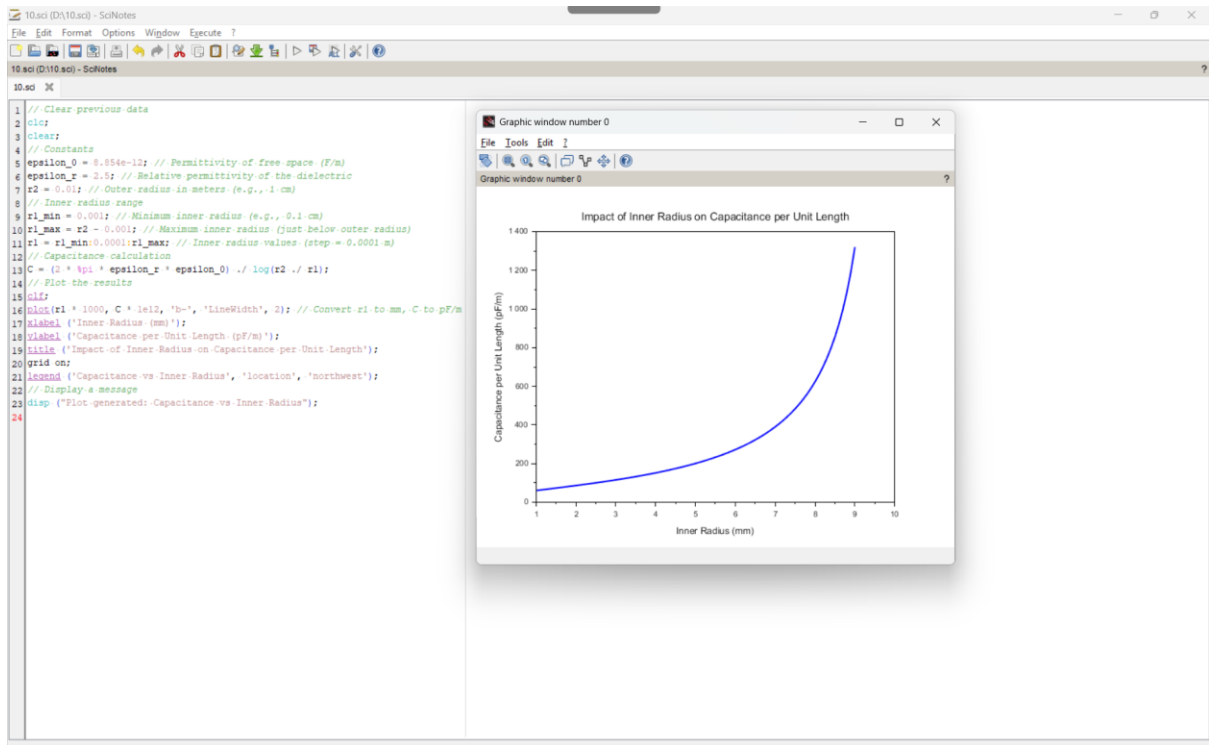
$$= \frac{222.413 \times 10^{-12}}{0.6931}$$

$$= 320.89 \times 10^{-12}$$

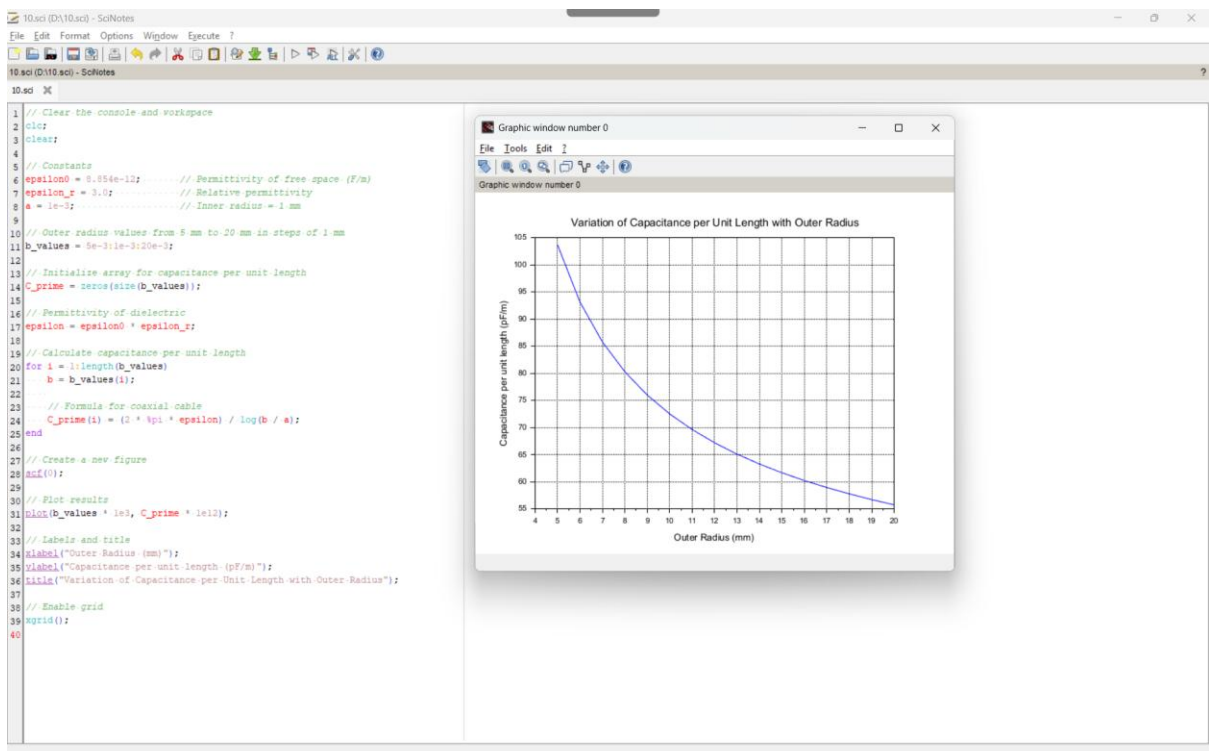
$$= 3.209 \times 10^{-10}$$

$$\therefore C = 3.209 \times 10^{-10} \text{ F}$$

CASE STUDY 1: Variation with Inner Radius



CASE STUDY 2: Variation with Outer Radius



CASE STUDY 3: Variation with Relative Permittivity

